
The Power of Catalytic Local Operations: Activation of Bell nonlocality and quantum state transformations

Jessica Bavaresco



Based on

Catalytic Activation of Bell Nonlocality

Jessica Bavaresco, Nicolas Brunner, Antoine Girardin, Pavel Sekatski, Patryk Lipka-Bartosik

[Phys. Rev. Lett. 135, 220203 \(2025\)](#), arXiv:2504.02042 [quant-ph]

and

The power of quantum catalytic local operations

Patryk Lipka-Bartosik, Jessica Bavaresco, Nicolas Brunner, Pavel Sekatski

[Phys. Rev. A 113, 062216 \(2026\)](#), arXiv:2510.08320 [quant-ph]



**UNIVERSITÉ
DE GENÈVE**

Catalysis

Catalysis

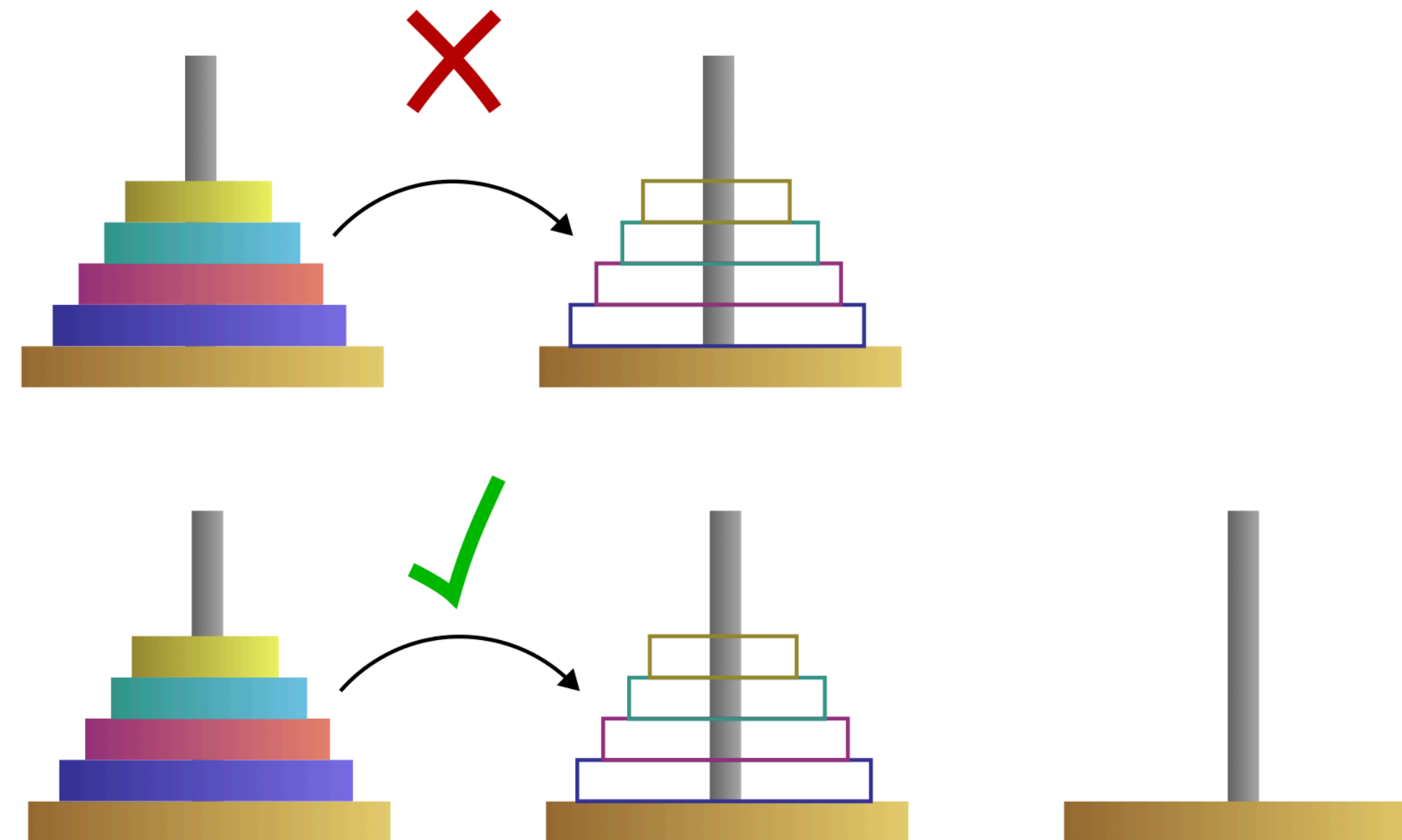


Image: P. Lipka-Bartosik, H. Wilming, N.H.Y. Ng, Rev. Mod. Phys. 96, 025005 (2024)

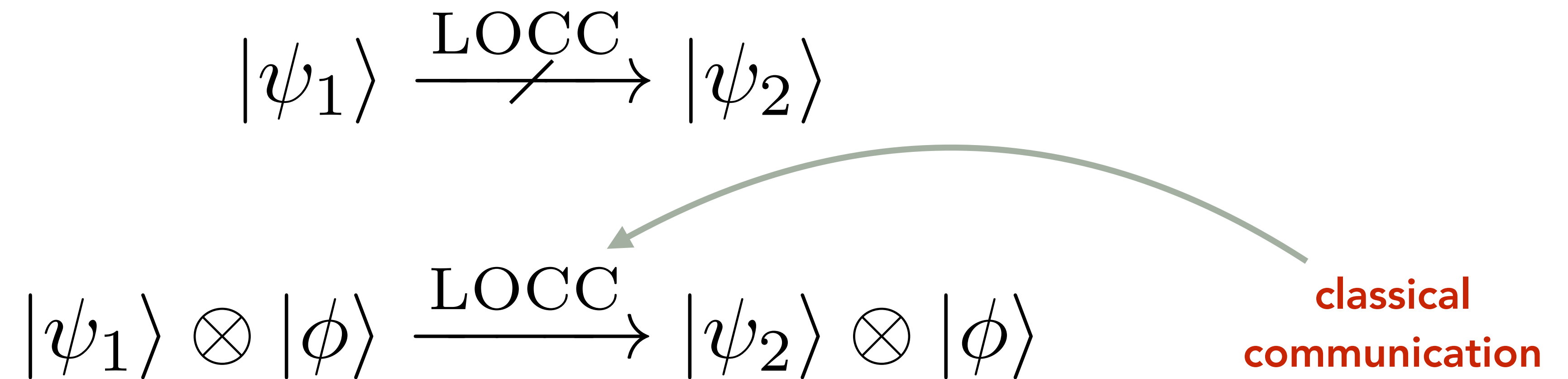
Quantum catalysis

$$|\psi_1\rangle \xrightarrow{\not\text{LOCC}} |\psi_2\rangle$$

$$|\psi_1\rangle \otimes |\phi\rangle \xrightarrow{\text{LOCC}} |\psi_2\rangle \otimes |\phi\rangle$$

D. Jonathan and M. B. Plenio, Phys. Rev. Lett 83, 3566 (1999)

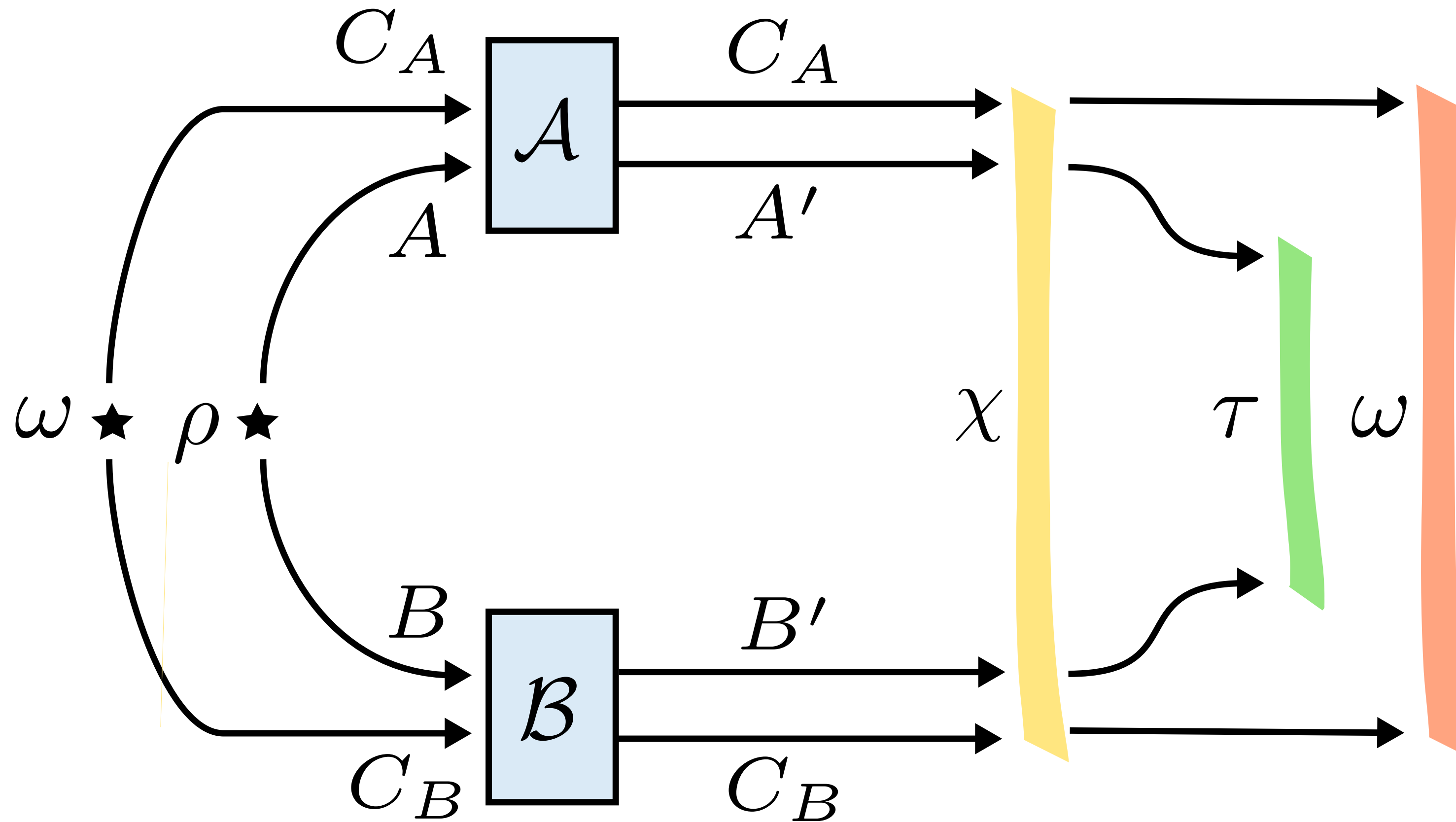
Quantum catalysis



D. Jonathan and M. B. Plenio, Phys. Rev. Lett 83, 3566 (1999)

We are interested in detangling the role of **classical communication** from that of **quantum catalysis** in state transformations.

Catalytic Local Operations



$$\chi_{A'B'C_AC_B} \neq \tau_{A'B'} \otimes \omega_{C_AC_B}$$

may be classically correlated
or entangled

First result: A Catalytic Transformation

$$\forall \rho_{AB} \in \mathcal{L}(\mathbb{C}^d \otimes \mathbb{C}^d), \quad \forall n \in \mathbb{N}$$

$$\rho_{AB} \xrightarrow{\text{CLO}} \tau_{A'B'} = \frac{1}{n} (\rho_{AB})^{\otimes n} \otimes [00]_{R_A R_B} + \frac{n-1}{n} (\sigma_A \otimes \sigma_B)^{\otimes n} \otimes [11]_{R_A R_B}$$

R. Duan, Y. Feng, X. Li, and M. Ying, Phys. Rev. A 71, 042319 (2005)

P. Lipka-Bartosik, P. Skrzypczyk, Phys. Rev. Lett. 127, 080502 (2021)

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deterministic
no shared randomness
no classical communication

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deterministic

no shared randomness

no classical communication

catalyst state


$$\omega_{C_A C_B} = \frac{1}{n} \sum_{i=0}^{n-1} (\rho_{\tilde{A}\tilde{B}})^{\otimes i} \otimes (\sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}})^{\otimes (n-1-i)} \otimes [ii]_{\tilde{R}_A \tilde{R}_B}$$

R. Duan, Y. Feng, X. Li, and M. Ying, Phys. Rev. A 71, 042319 (2005)

P. Lipka-Bartosik, P. Skrzypczyk, Phys. Rev. Lett. 127, 080502 (2021)

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probabilistic cloning?

No!

catalyst not universal


$$\omega_{C_A C_B} = \frac{1}{n} \sum_{i=0}^{n-1} (\rho_{\tilde{A}\tilde{B}})^{\otimes i} \otimes (\sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}})^{\otimes (n-1-i)} \otimes [ii]_{\tilde{R}_A \tilde{R}_B}$$

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JB, N. Brunner, A. Girardin, P. Sekatski, P. Lipka-Bartosik, Phys. Rev. Lett. 135, 220203 (2025)

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A} \tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

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1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

2) If in branch: $[00]_{\tilde{R}_A \tilde{R}_B}$ $\rho_{AB} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B}$

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

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2) If in branch: $[00]_{\tilde{R}_A \tilde{R}_B}$

$$\rho_{AB} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B}$$

(i) Swap systems $A \leftrightarrow \tilde{A}$ and $B \leftrightarrow \tilde{B}$

$$\mapsto \sigma_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B}$$

(ii) Flip classical registers $[00]_{\tilde{R}_A \tilde{R}_B} \mapsto [11]_{\tilde{R}_A \tilde{R}_B}$

$$\mapsto \sigma_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$$

(iii) Locally prepare system $A'B' = A_1 B_1 A_2 B_2$
in product state $\sigma_{A_1 B_1} \otimes \sigma_{A_2 B_2}$

$$\mapsto \sigma_{A_1 B_1} \otimes \sigma_{A_2 B_2} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$$

(iv) Prepare new classical registers $[11]_{R_A R_B}$

$$\mapsto \sigma_{A_1 B_1} \otimes \sigma_{A_2 B_2} \otimes [11]_{R_A R_B} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$$

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

2) If in branch: $[11]_{\tilde{R}_A \tilde{R}_B}$ $\rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$

Proof: A Catalytic Transformation

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1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

2) If in branch: $[11]_{\tilde{R}_A \tilde{R}_B}$ $\rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$

(i) Locally prepare system $A'B' = A_1 B_1 A_2 B_2$
in state $\rho_{A_1 B_1} \otimes \rho_{A_2 B_2}$ and system $\tilde{A}\tilde{B}$
in product state $\sigma_{\tilde{A}\tilde{B}}$

$$\mapsto \rho_{A_1 B_1} \otimes \rho_{A_2 B_2} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}$$

(ii) Flip classical registers $[11]_{\tilde{R}_A \tilde{R}_B} \mapsto [00]_{\tilde{R}_A \tilde{R}_B}$

$$\mapsto \rho_{A_1 B_1} \otimes \rho_{A_2 B_2} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B}$$

(iii) Prepare new classical registers $[00]_{R_A R_B}$

$$\mapsto \rho_{A_1 B_1} \otimes \rho_{A_2 B_2} \otimes [00]_{R_A R_B} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B}$$

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

- 1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$
- 2) Perform local operations

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

2) Perform local operations

$$\begin{aligned} \text{3) Resulting state: } \chi_{A'B'C_A C_B} = \frac{1}{2} & \left(\rho_{A_1 B_1} \otimes \rho_{A_2 B_2} \otimes [00]_{R_A R_B} \otimes \sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \right. \\ & \left. \sigma_{A_1 B_1} \otimes \sigma_{A_2 B_2} \otimes [11]_{R_A R_B} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right) \end{aligned}$$

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

1) Alice and Bob measure their local classical register $\tilde{R}_A \tilde{R}_B$

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Marginals:

$$\text{tr}_{A'B'}(\chi_{A'B'C_A C_B}) = \frac{1}{2} (\sigma_{\tilde{A}\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B}) = \omega_{C_A C_B} \quad \checkmark$$

Catalyst preserved!

Proof: A Catalytic Transformation

$$n = 2 \quad \rho_{AB} \otimes \omega_{C_A C_B} = \frac{1}{2} \left(\rho_{AB} \otimes \sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}} \otimes [00]_{\tilde{R}_A \tilde{R}_B} + \rho_{AB} \otimes \rho_{\tilde{A}\tilde{B}} \otimes [11]_{\tilde{R}_A \tilde{R}_B} \right)$$

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Final state:

$$\text{tr}_{C_A C_B} (\chi_{A'B'C_A C_B}) = \frac{1}{2} (\rho_{AB}^{\otimes 2} \otimes [00]_{R_A R_B} + (\sigma_A \otimes \sigma_B)^{\otimes 2} \otimes [11]_{R_A R_B}) = \tau_{A'B'} \checkmark$$

Target state achieved!

First result: A Catalytic Transformation

$$\forall \rho_{AB} \in \mathcal{L}(\mathbb{C}^d \otimes \mathbb{C}^d), \quad \forall n \in \mathbb{N}$$

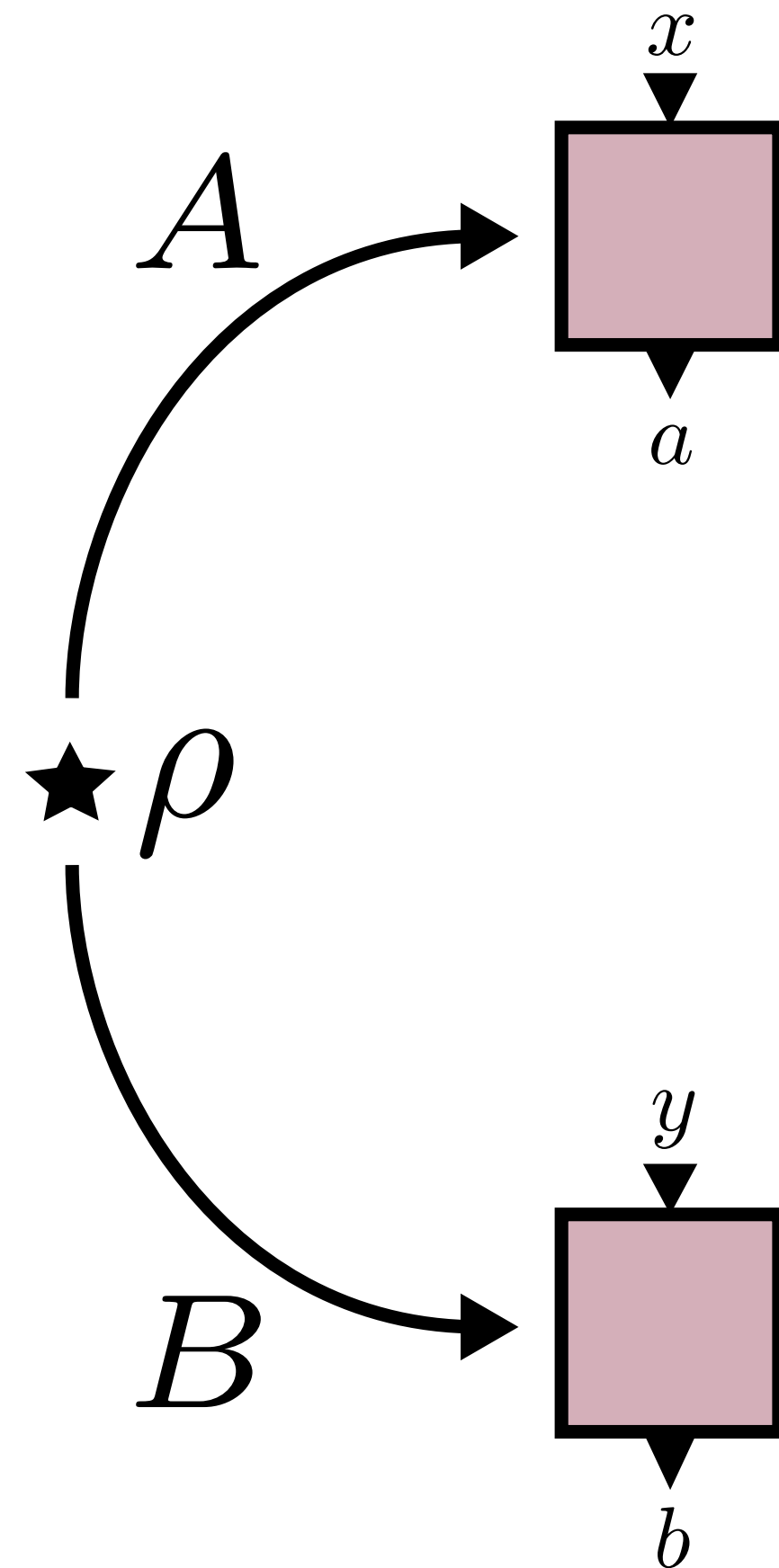
$$\rho_{AB} \xrightarrow{\text{CLO}} \tau_{A'B'} = \frac{1}{n} (\rho_{AB})^{\otimes n} \otimes [00]_{R_A R_B} + \frac{n-1}{n} (\sigma_A \otimes \sigma_B)^{\otimes n} \otimes [11]_{R_A R_B}$$

catalyst state 

$$\omega_{C_A C_B} = \frac{1}{n} \sum_{i=0}^{n-1} (\rho_{\tilde{A}\tilde{B}})^{\otimes i} \otimes (\sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}})^{\otimes (n-1-i)} \otimes [ii]_{\tilde{R}_A \tilde{R}_B}$$

Activation of Bell Nonlocality

Bell-local quantum states



Bell scenario

Bell-local states: A bipartite quantum state ρ_{AB} is called Bell local if, for all sets of local measurements $\{M_{a|x}^A\}$ and $\{M_{b|y}^B\}$, the correlations

$$p(ab|xy) = \text{tr}[(M_{a|x}^A \otimes M_{b|y}^B)\rho_{AB}]$$

admit a local hidden variable model, according to

$$p(ab|xy) = \sum_{\lambda} \pi(\lambda) p_A(a|x, \lambda) p_B(b|y, \lambda).$$

Activation of Bell nonlocality

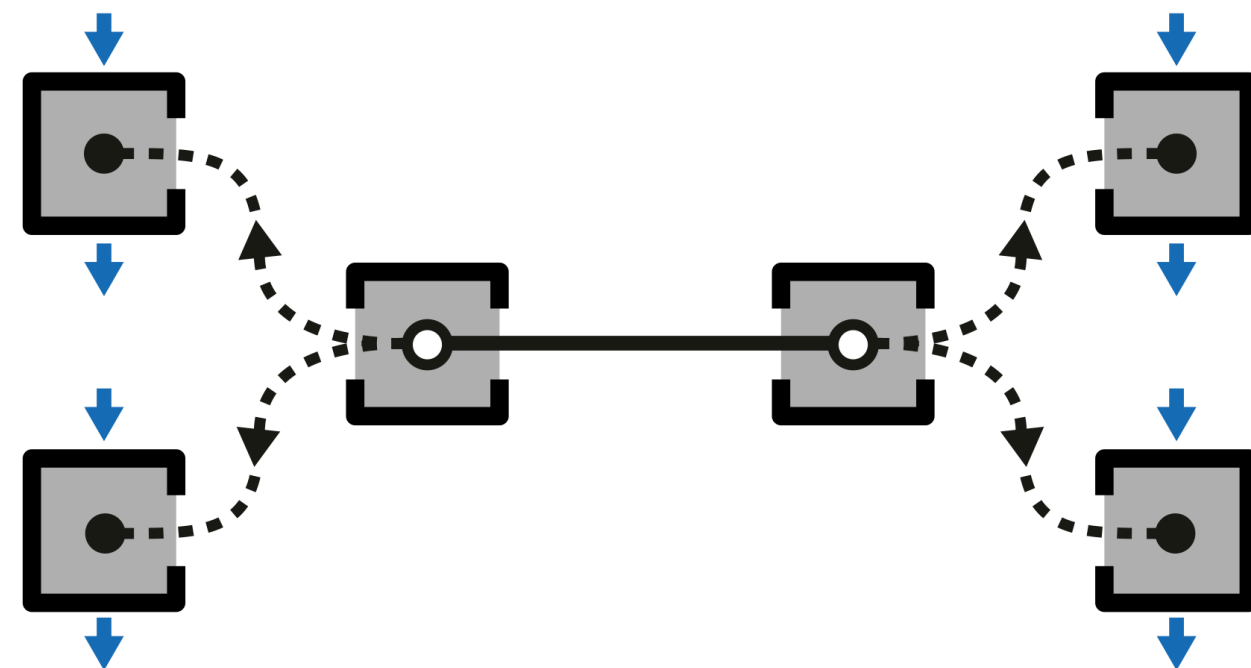
Filtering (SLOCC, “hidden nonlocality”)

S. Popescu, Phys. Rev. Lett. 74, 2619 (1995)
N. Gisin, Phys. Lett. A 210, 151 (1996)

...

F. Hirsch, M.T. Quintino, J. Bowles, N. Brunner,
Phys. Rev. Lett. 111, 160402 (2013)

...

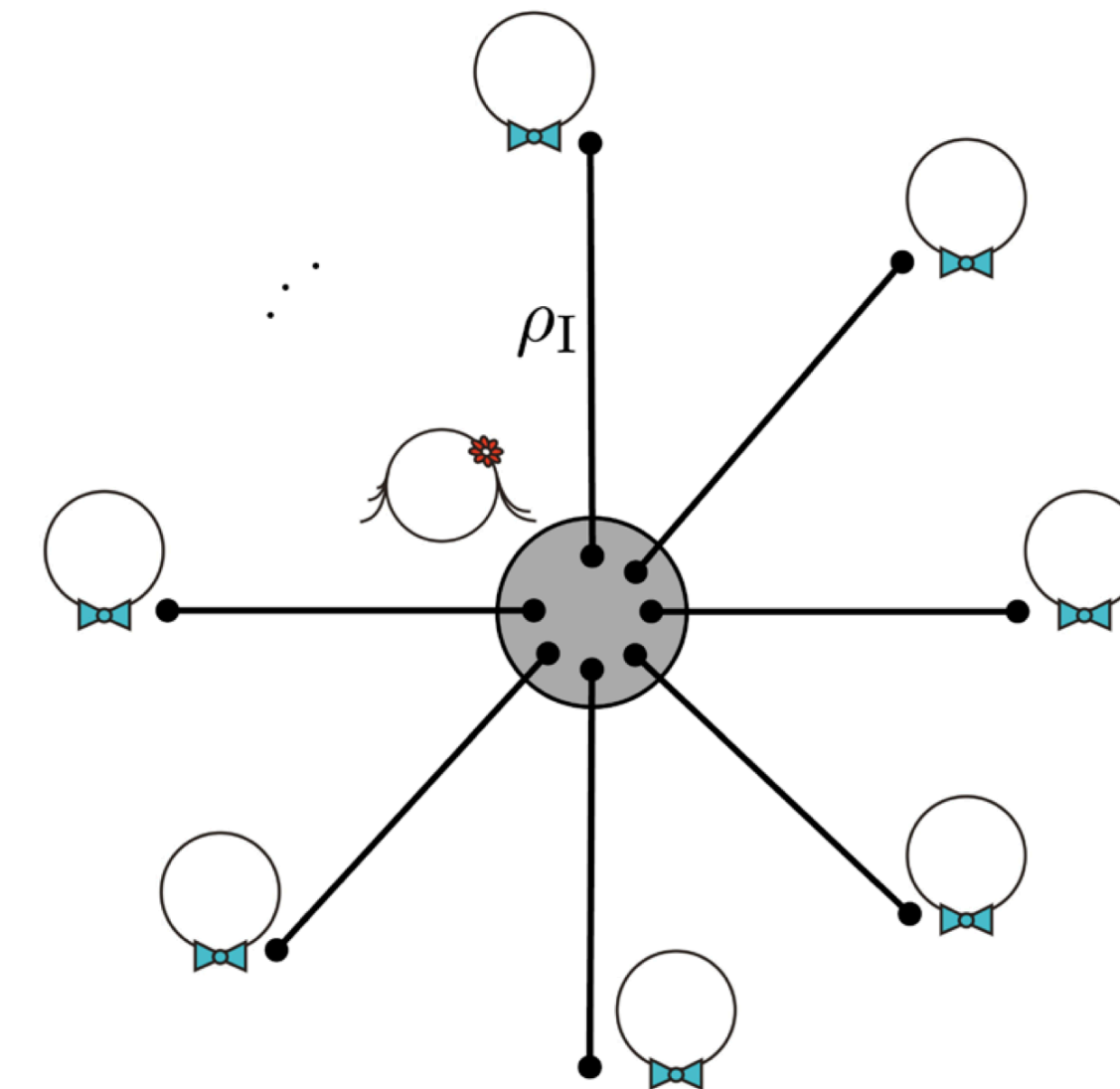
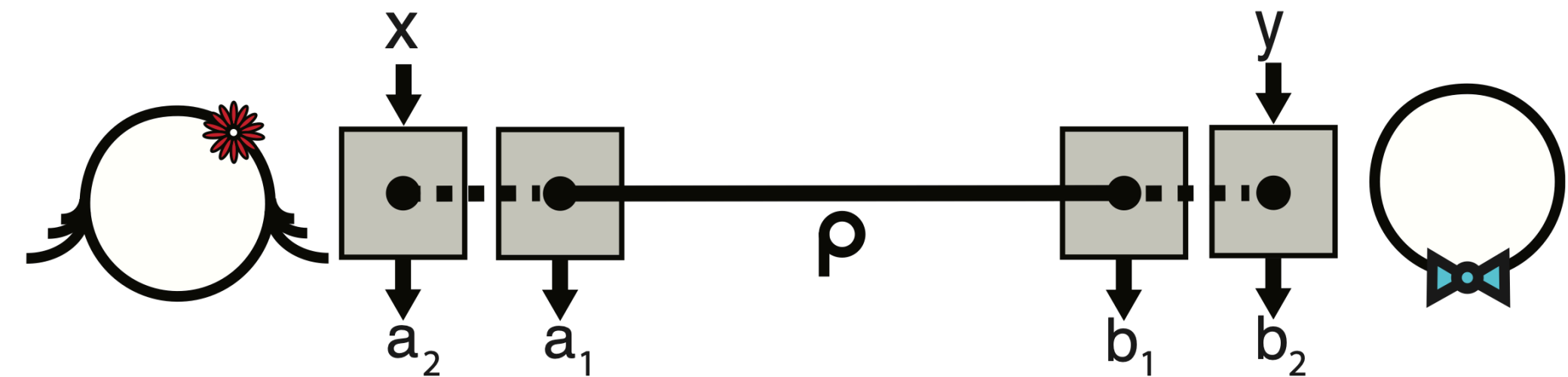


Broadcasting

J. Bowles, F. Hirsch, D. Cavalcanti, Quantum 5, 499 (2021)

...

J. Pauwels, P. Sekatski, arXiv:2512.15656 [quant-ph]

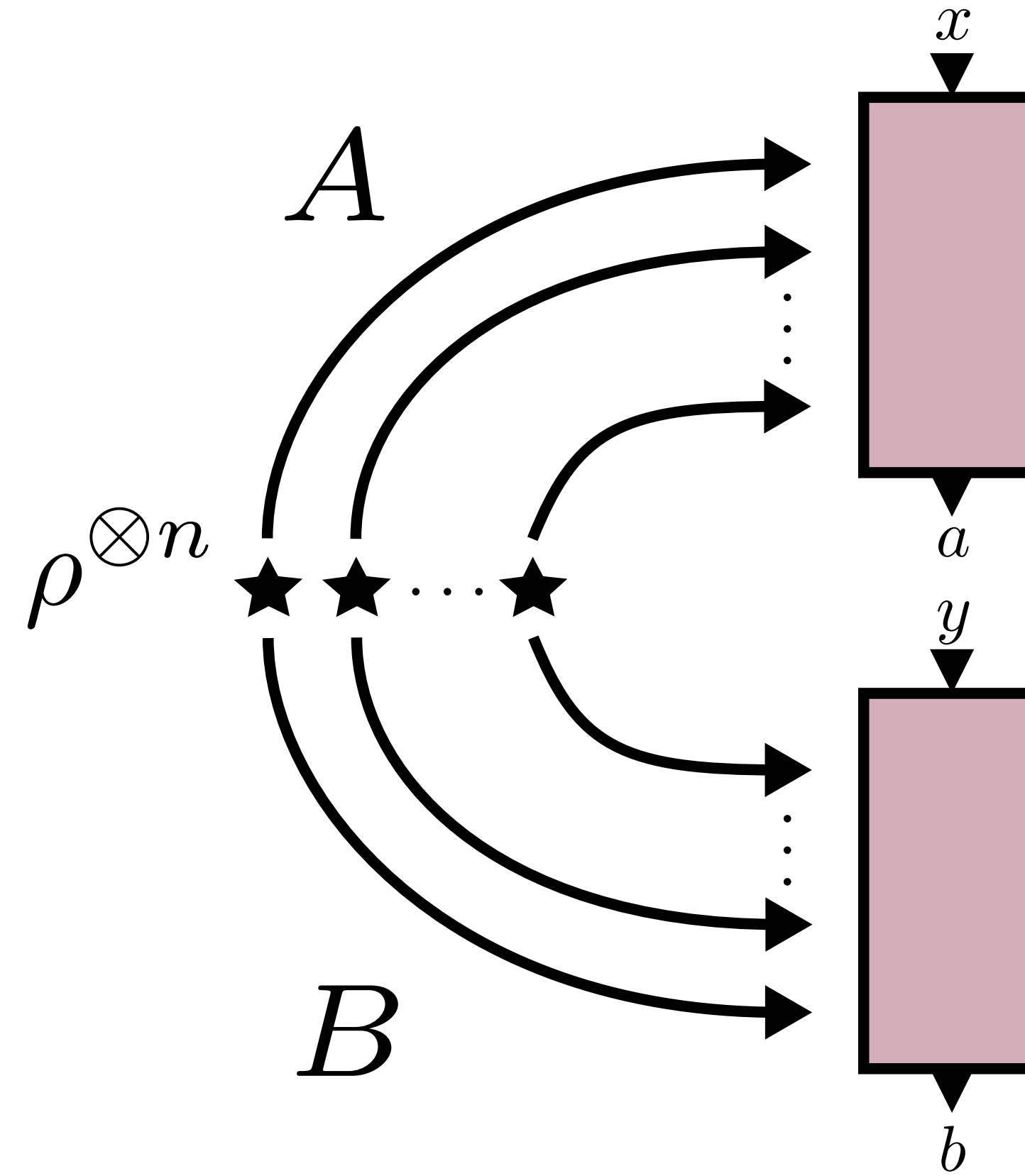


Networks

D. Cavalcanti, M. L. Almeida, V. Scarani, A. Acin,
Nat. Commun. 2, 184 (2011)

...

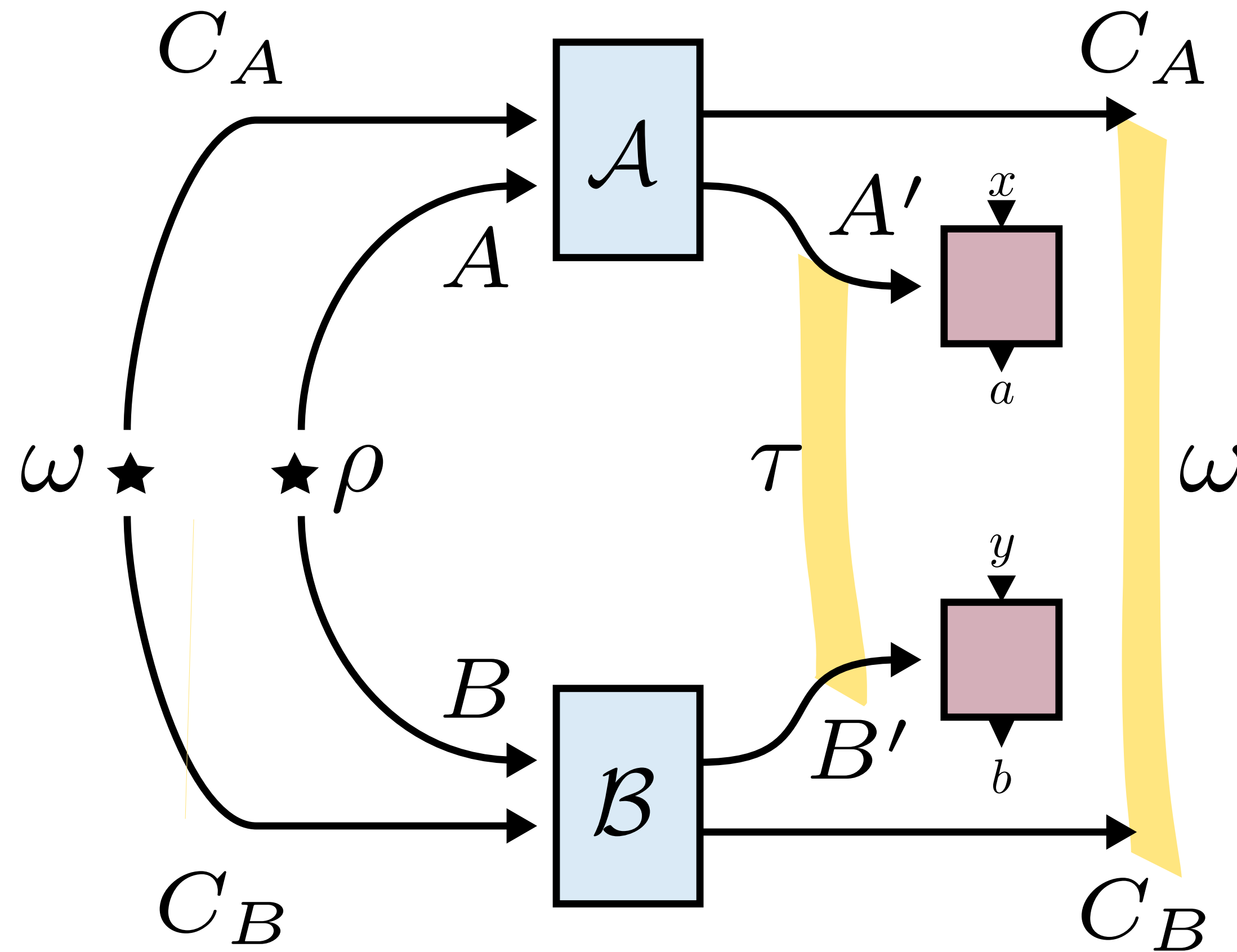
Activation of Bell nonlocality



Many-copy

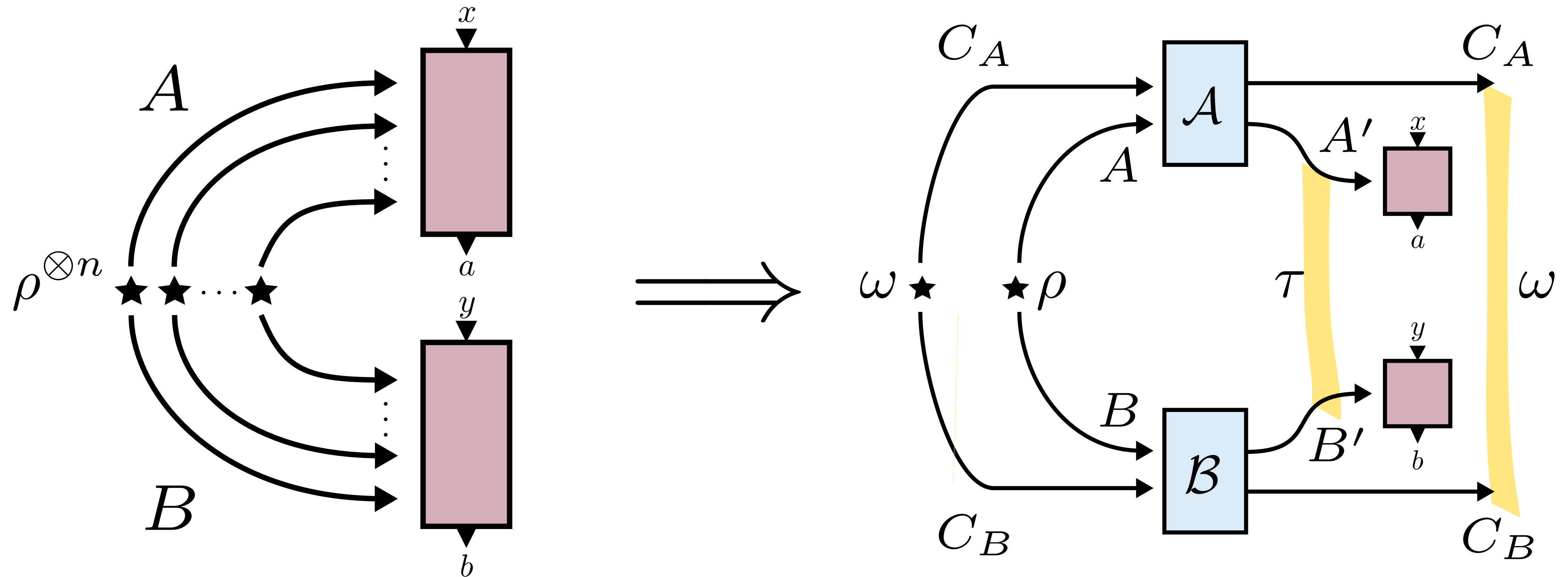
C. Palazuelos, Phys. Rev. Lett. 109, 190401 (2012)

Second result: Catalytic Activation of Bell NL



Theorem 1. Catalytic activation of Bell NL is possible.

Second result: Catalytic Activation of Bell NL



Theorem 1. In particular, many-copy activation implies catalytic activation.

Second result: Catalytic Activation of Bell NL

Observation 1. If $\rho_{AB}^{\otimes n}$ is Bell nonlocal, $\tau_{A'B'}$ is also Bell nonlocal.

$$\rho_{AB} \xrightarrow{\text{CLO}} \tau_{A'B'} = \frac{1}{n} (\rho_{AB})^{\otimes n} \otimes [00]_{R_A R_B} + \frac{n-1}{n} (\sigma_A \otimes \sigma_B)^{\otimes n} \otimes [11]_{R_A R_B}$$

If the NL of a Bell-local state is activatable with n copies,
it can also be catalytically activatable, “consuming” only 1 copy.

Second result: Catalytic Activation of Bell NL

Observation 1. If $\rho_{AB}^{\otimes n}$ is Bell nonlocal, $\tau_{A'B'}$ is also Bell nonlocal.

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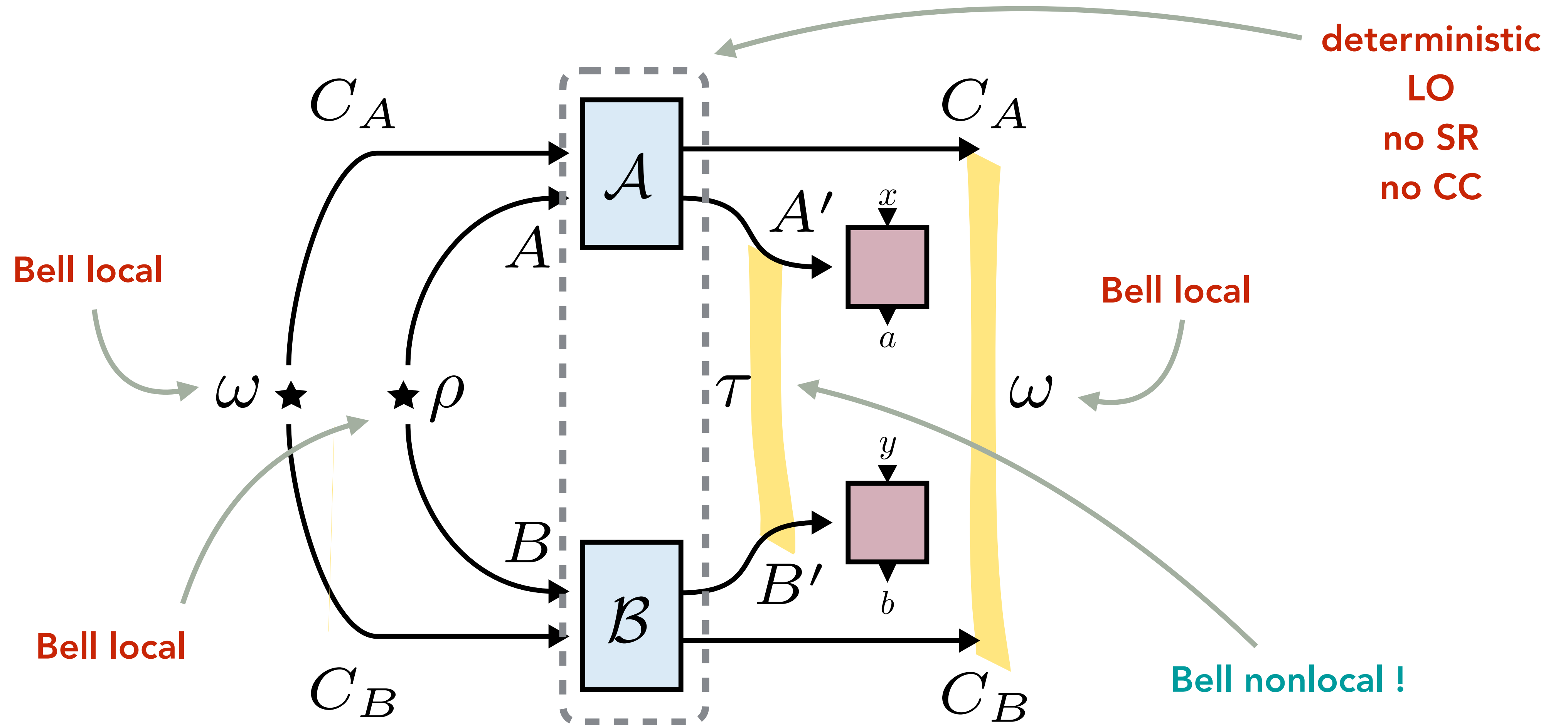
If the NL of a Bell-local state is activatable with n copies,
it can also be catalytically activatable, “consuming” only 1 copy.

Observation 2. If $\rho_{AB}^{\otimes (n-1)}$ is Bell local, $\omega_{C_A C_B}$ is also Bell local.

$$\omega_{C_A C_B} = \frac{1}{n} \sum_{i=0}^{n-1} (\rho_{\tilde{A}\tilde{B}})^{\otimes i} \otimes (\sigma_{\tilde{A}} \otimes \sigma_{\tilde{B}})^{\otimes (n-1-i)} \otimes [ii]_{\tilde{R}_A \tilde{R}_B}$$

Catalytic activation of Bell NL is possible even with a **local catalyst!**

Second result: Catalytic Activation of Bell NL



Catalytic **super**-activation of Bell NL is possible.

Quantum state transformations

CLO(d)

CLO(d): Catalytic Local Operations. The class of dimension-bounded catalytic local operations contains all transformations of a bipartite state into another that can be implemented via local CPTP maps with the assistance of a catalyst state whose marginal is preserved and which has Schmidt number respecting

$$\text{SN}(\omega_{C_A C_B}) \leq d .$$

SLOCCQ(d)

SLOCCQ(d): Stochastic Local Operations with Classical Communication and Quantum communication. This class of transformations contains all sequences of stochastic local operations, classical communication, and dimension-bounded quantum communication, where the dimension of the quantum communication is bounded in the following way. Let d_i be the dimension of the quantum system sent from one party to another in round i . We require that

$$\prod_i d_i \leq d .$$

State transformations

Observation 3. $\text{CLO}^{(d)} \subseteq \text{LOSE}^{(d)} \subseteq \text{SLOCCQ}^{(d)}$

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Observation 5. $\text{LOCC} \not\subseteq \text{CLO}^{(d)}$

State transformations

Observation 3. $\text{CLO}^{(d)} \subseteq \text{LOSE}^{(d)} \subseteq \text{SLOCCQ}^{(d)}$

Observation 4. $\text{CLO}^{(1)} = \text{LOSR}$

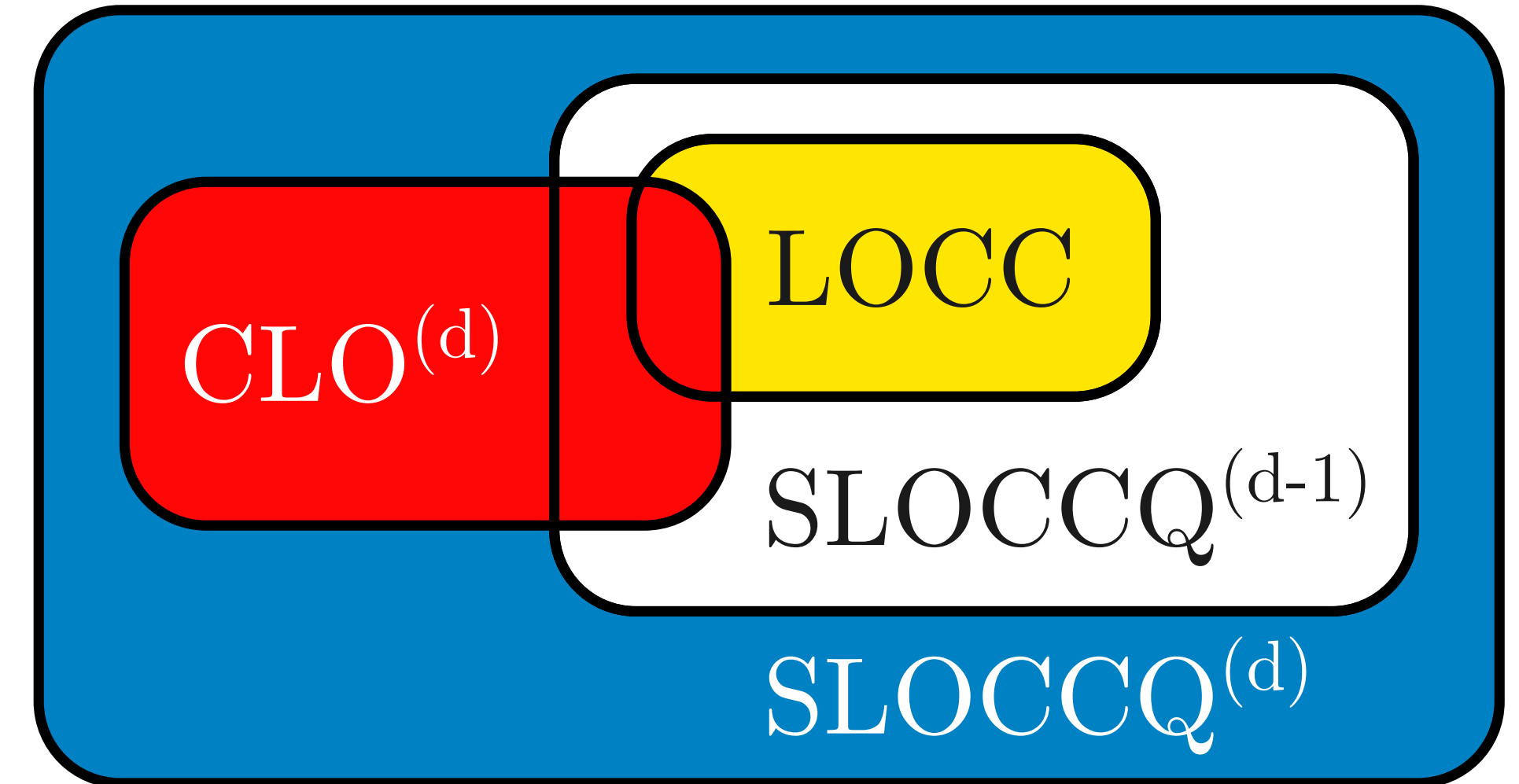
Observation 5. $\text{LOCC} \not\subseteq \text{CLO}^{(d)}$

$$\rho_{ABC} = \sum_{i=0}^1 \frac{1}{2} [0]_A \otimes [i]_B \otimes [i]_C \xrightarrow[\text{LOSE}]{\text{LOCC}} \sigma_{ABR} = \sum_{i=0}^1 \frac{1}{2} [i]_A \otimes [0]_B \otimes [i]_C$$

Third result: State transformations

Theorem 2.

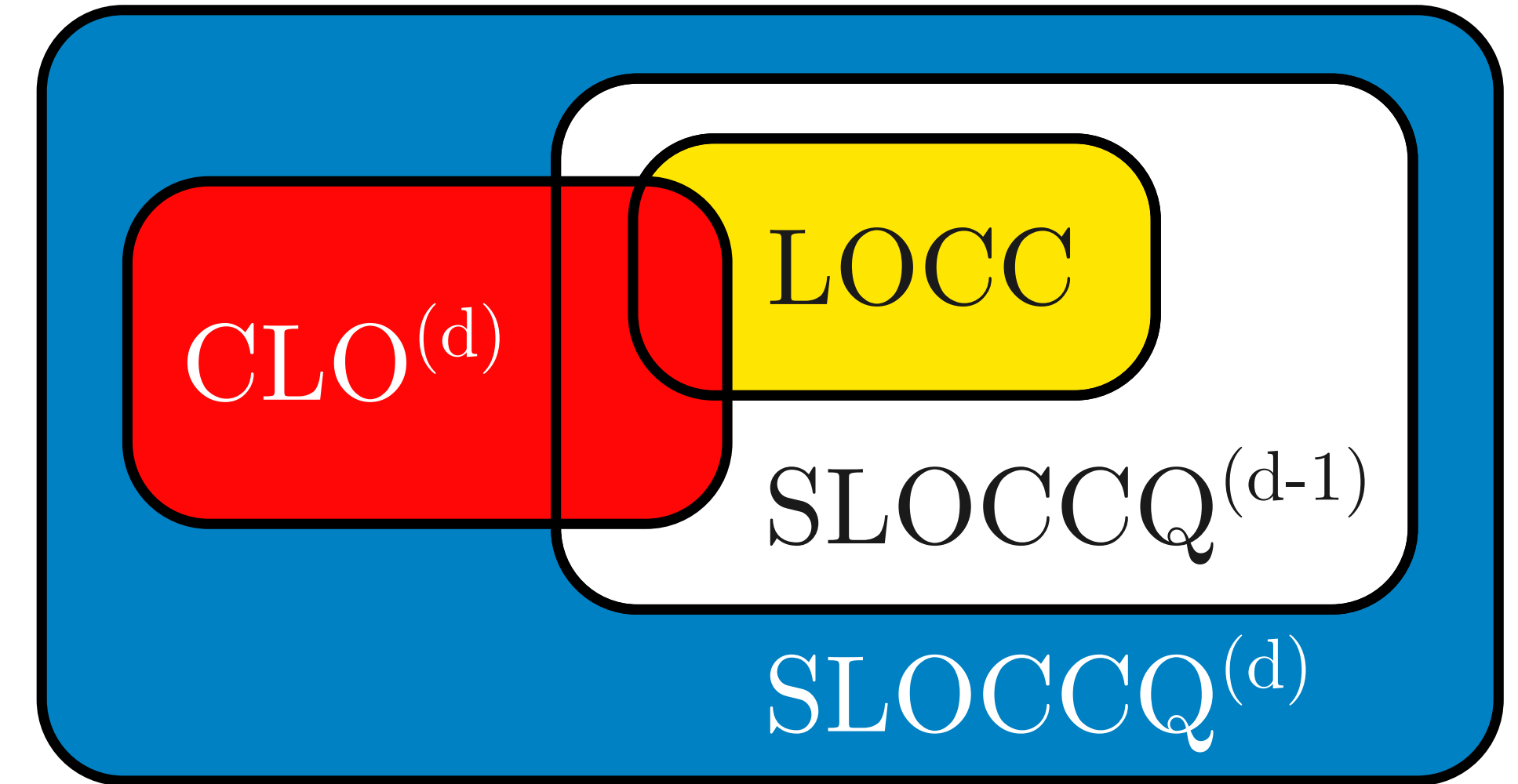
$$\text{CLO}^{(d)} \not\subset \text{SLOCCQ}^{(d-1)}$$



Third result: State transformations

Theorem 2.

$$\text{CLO}^{(d)} \not\subset \text{SLOCCQ}^{(d-1)}$$



Take $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$ with $\text{SN}(\rho_{AB}) = 2$.

Via $\text{CLO}^{(2^n)}$ it can be transformed into $\tau_{A'B'}$ with $\text{SN}(\tau_{A'B'}) = 2^{n+1}$.

$$\frac{\text{SN}(\rho_{AB})}{\text{SN}(\tau_{A'B'})} = 2^n > d = 2^n - 1 \implies \text{not possible with } \text{SLOCCQ}^{(2^n-1)}$$

Summary

Lemma 1. $\rho_{AB} \xrightarrow{\text{CLO}} \tau_{A'B'} = \frac{1}{n} (\rho_{AB})^{\otimes n} \otimes [00]_{R_A R_B} + \frac{n-1}{n} (\sigma_A \otimes \sigma_B)^{\otimes n} \otimes [11]_{R_A R_B}$

Observation 1. If $\rho_{AB}^{\otimes n}$ is Bell nonlocal, $\tau_{A'B'}$ is also Bell nonlocal.

Observation 2. If $\rho_{AB}^{\otimes(n-1)}$ is Bell local, $\omega_{C_A C_B}$ is also Bell local.

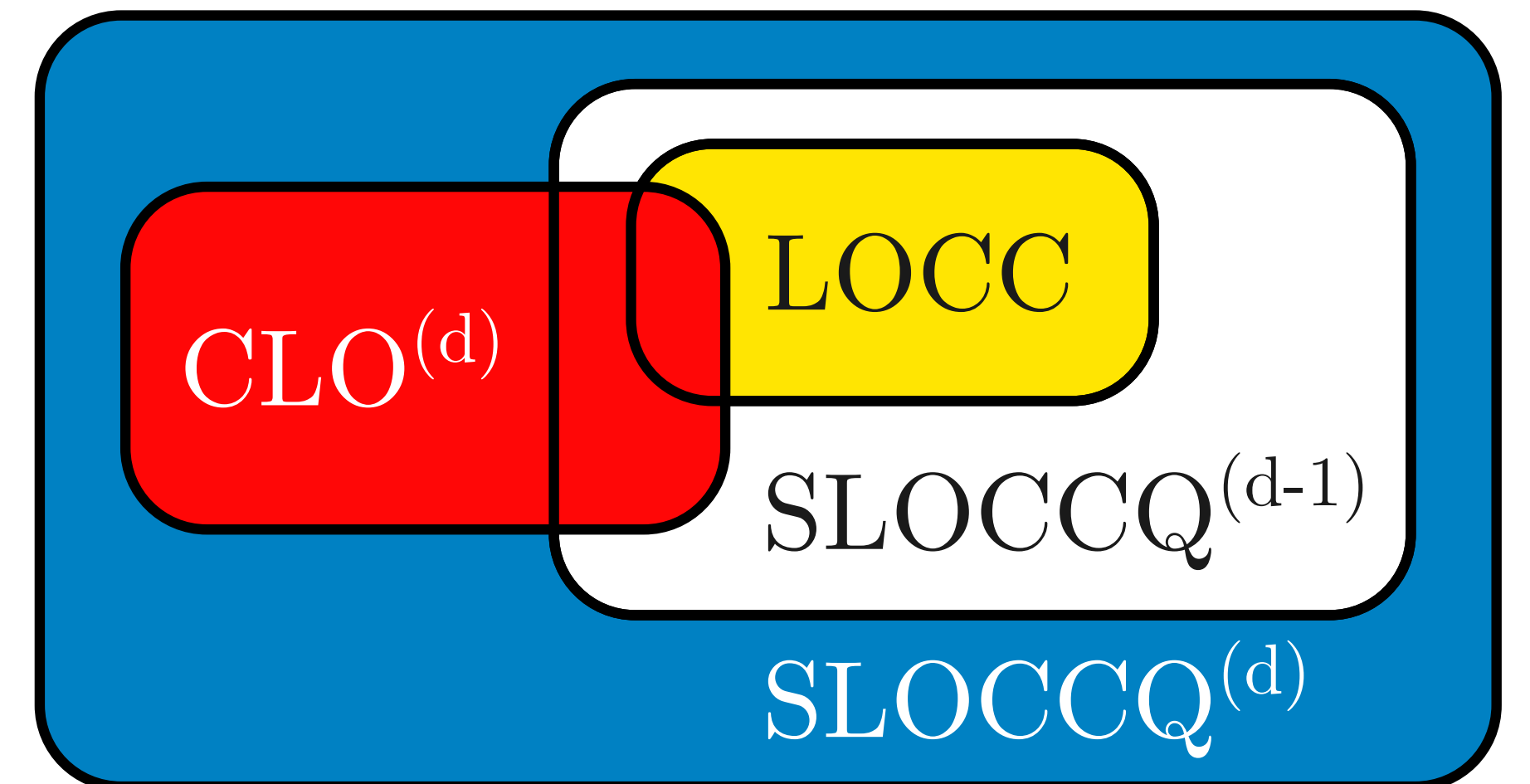
Theorem 1. Many-copy activation implies catalytic activation with a local catalyst.

Observation 3. $\text{CLO}^{(d)} \subseteq \text{LOSE}^{(d)} \subseteq \text{SLOCCQ}^{(d)}$

Observation 4. $\text{CLO}^{(1)} = \text{LOSR}$

Observation 5. $\text{LOCC} \not\subseteq \text{CLO}^{(d)}$

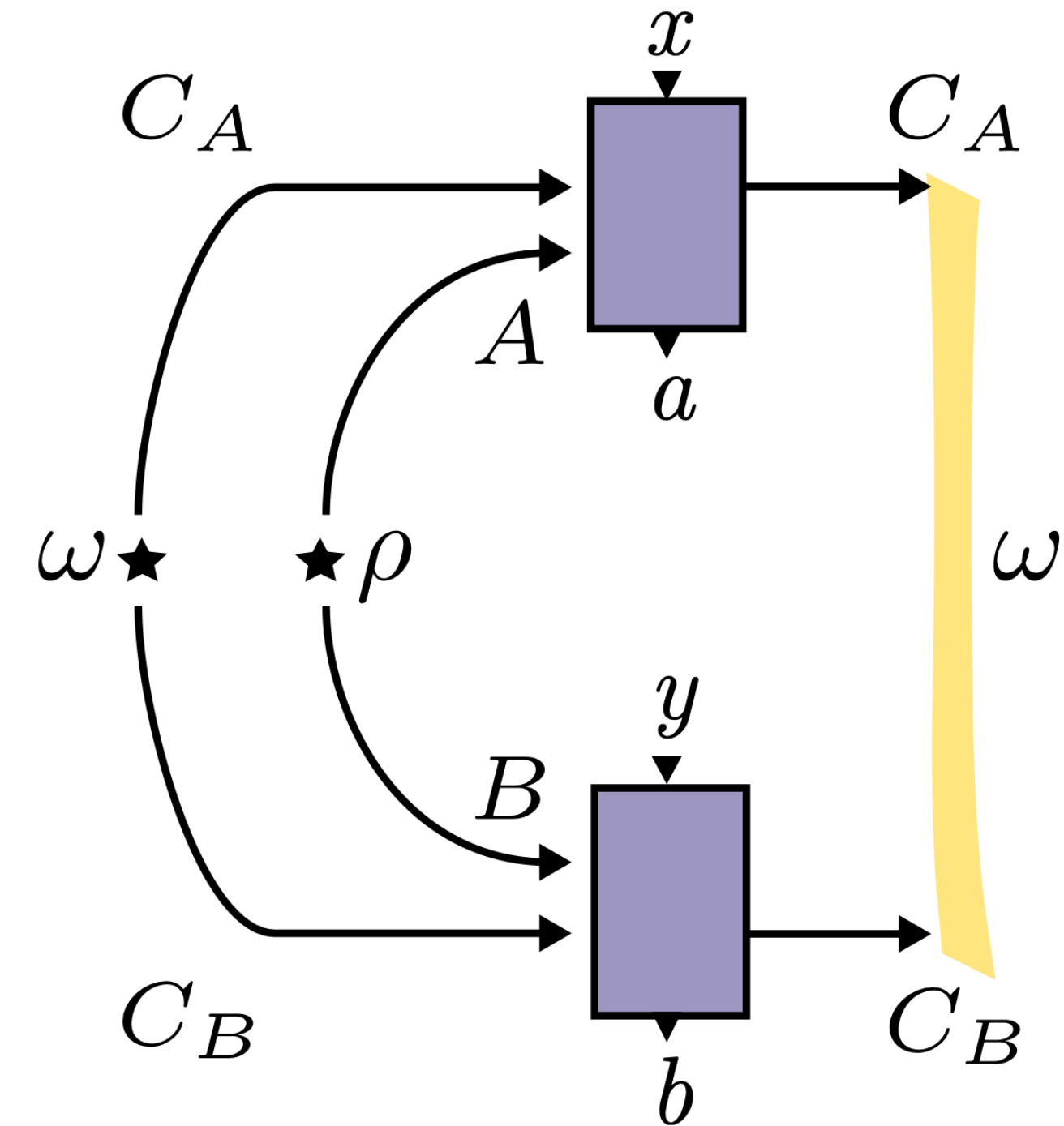
Theorem 2. $\text{CLO}^{(d)} \not\subseteq \text{SLOCCQ}^{(d-1)}$



Future directions

- Bell nonlocality

- **State activation:** can quantum states be catalytically activated even when they cannot be many-copy activatable?
- **Other forms of catalysis:** can other quantum states be activated in different catalysis activation scenarios?



- Entanglement theory

- **Monotones:** which entanglement measures are monotonic under CLO and which (such as the Schmidt number) are not?

Thank you!